



Reserved, Not Shortened: Designing and Evaluating a Paper-Slip Reservation Scheme for the Shared Staff Microwave

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Abstract. Complaints to Facilities about the shared microwave in the School of Continuous Improvement's ground-floor break room had, for three consecutive months, named "kitchen-amenity fairness" among the service log's most frequent categories, averaging 13.7 tickets a month across roughly ninety staff sharing a single appliance. In response, Facilities designed and deployed a paper-slip reservation scheme — a wall-mounted board on which staff write their name against a ten-minute slot, with one slot in four set aside as a Quick-Reheat Lane for reheats advertised under ninety seconds. This paper reports the scheme's design and a same-cohort evaluation of its two candidate outcomes. Complaint volume — the measure the service log tracks, and the one Facilities' continuation report credits the scheme with fixing — fell by four-fifths across the two quarters following deployment. A direct field measurement of wait-to-reheat time, stopwatched from arrival to reheat start across matched lunch-hour samples before and after deployment, finds no detectable difference between the two periods ($n = 64, 61$; n.s.). An ablation removing the Quick-Reheat Lane for a further quarter moves neither outcome outside the range already observed. The slip, we suggest, functions chiefly as a fairness-legible artefact the service log can point to, rather than as the mechanism actually governing how long anyone waits — a distinction the continuation report has had no occasion to draw.

Introduction

Slop University's largest single-microwave break room sits on the ground floor of the School of Continuous Improvement's main wing, serving close to ninety professional and academic staff across three floors on an unscheduled, first-come basis. The room is also where colleagues who might not otherwise cross paths do — shared workplace meals are documented to carry organisational benefits distinct from anything a fairness ticket measures [1] — but the ninety staff who share it are, for this paper's purposes, its population of record. For as long as Facilities' service log has broken kitchen-amenity tickets into subcategories, fairness has been the largest of them: staff queuing behind a colleague's frozen lasagne, a departmental team reheating four trays in sequence, the same handful of names recurring in informal

complaints to line managers who had no formal channel to raise them through.

Facilities' response, designed and trialled over three quarters, was a paper-slip reservation scheme: a laminated board beside the microwave on which any staff member may write their name against one of six ten-minute slots per lunch hour, with a Quick-Reheat Lane — one slot in four — reserved for reheats advertised at under ninety seconds and released back to the general roster if the claimant runs over. The scheme requires no app, no login, and no ongoing Facilities administration beyond a weekly slip changeover; it is exactly the kind of low-cost governance instrument an institution can point to as evidence a fairness complaint was heard.

Pointing to an instrument is not the same as measuring what it does. A published mechanism

that visibly responds to a complaint category can retire the complaint without necessarily retiring the underlying wait — the two are distinct claims, and a service log built to count tickets is well suited to the first and poorly suited to the second. This paper reports both counts for the same scheme, deployment, and time window, evaluated independently for the population it governs.

The paper's contributions, each modest by design:

1. We describe the design and deployment of a paper-slip reservation scheme, including the Quick-Reheat Lane, and report the sharp fall in complaint volume the scheme's continuation report credits it with.
2. We report a direct field measurement of wait-to-reheat time across matched pre- and post-deployment samples, finding no detectable change against the same population and window the complaint count covers.
3. We report an ablation removing the Quick-Reheat Lane for a further quarter, finding it moves neither outcome measurably, a result we retain rather than treat as a design failure.

The scheme is not shown to be worthless, and this paper does not attempt to show that; it is shown to be, on the wait-time evidence available, no faster than the arrangement it replaced.

Related work

Institutional measures that come to stand in for the outcome they were designed to serve are a documented pathology, not an unusual one: a quantitative indicator adopted for social decision-making is argued to distort the process it monitors [2], and optimisation pressure is shown to separate a target from the substantive goal it once tracked in settings from academic publishing metrics [3] to targets-and-terror regimes in public health [4]. Complaint counts carry a long-known undercount: registered complaints are argued to show only the tip of a larger annoyance iceberg [5], a systematic share of dissatisfied customers is documented never to voice a complaint at all [6], and nonbehavioural withdrawal is shown to be a common alternative to voicing dissatisfaction outright [7].

Reservation mechanisms and the queues they replace have their own comparative literature: reservation and queueing systems are shown to be substitutable only under specific load

conditions [8], segmenting customers into priority classes carries a perceived-fairness cost even where it raises aggregate throughput [9], and formally equitable priority queues are documented to produce felt inequity in the class not prioritised [10]. Queue behaviour absent a formal mechanism is itself shown to self-organise around locally observed norms [11], and ethnographic accounts of queue-jumping find informal negotiation around a formal rule to be a durable, largely autonomous practice [12]. Measurement is not a neutral bystander: the foundational claim that observation changes observed behaviour is shown, on reanalysis of the original data, to be more equivocal than its folklore version [13], self-reported and directly measured behaviour are shown to diverge systematically [14], and formal voice channels are shown to substitute imperfectly for the exit or loyalty signals of underlying decline [15].

Prior work at this institution has traced a comparable divergence between an activation-based facilities metric and the behaviour it proxies, in a hydration-station tally [16] and a hand-sanitiser refill log [17]; the present case substitutes a complaint count for an activation count as the diverging proxy.

Method

Site and scheme design

The break room under study serves roughly ninety academic and professional staff across three floors of the School of Continuous Improvement's main wing from a single microwave, a ratio Facilities' own space-planning guidance treats as within tolerance for a single appliance. Before deployment, staff queued informally at the appliance with no signage, scheduling, or formal escalation path beyond a complaint to a line manager.

The Reheat Slot Roster, designed by Facilities over one quarter and deployed for two further quarters, replaces informal queuing with a laminated board mounted beside the microwave, ruled into six ten-minute slots per lunch hour (12:00–13:30); any staff member may claim a slot by writing their name and floor against it, with no digital record kept beyond the physical slip itself.

Parameter	Value	Basis
Slot length	10 min	Design brief
Slots / lunch hour	6	90 min / 10 min
Quick-Reheat share	1 in 4	Design brief
Quick-Reheat ceiling	90 s	On the board

A formal definition of fairness for divisible shared resources holds that no user has a justified complaint provided they receive either everything requested or their full entitlement of some bottleneck resource [18]; the Roster does not attempt this standard formally, approximating it instead with a fixed slot grid and the Quick-Reheat Lane as its sole differentiated class.

Measurement

Two independent measures were recorded across the same population and calendar window. Complaint volume is Facilities' own service-log count of tickets filed under the kitchen-amenity fairness subcategory, for the three months preceding deployment and the two quarters following it. Wait-to-reheat time is a direct field measurement: two observers, rostered to avoid the claimant's own knowledge of the sampling schedule, recorded the interval between a staff member's arrival at the appliance queue and the start of their reheat cycle, across matched weekday lunch-hour samples drawn before ($n = 64$) and after ($n = 61$) deployment.

Ablation

For one further quarter, the Quick-Reheat Lane was withdrawn from the board without other change to the scheme, restoring a uniform six-slot roster. Both complaint volume and wait-to-reheat time were recorded across this quarter ($n = 58$ wait-time observations) on the same protocol, for comparison against the two quarters immediately preceding the withdrawal.

Results

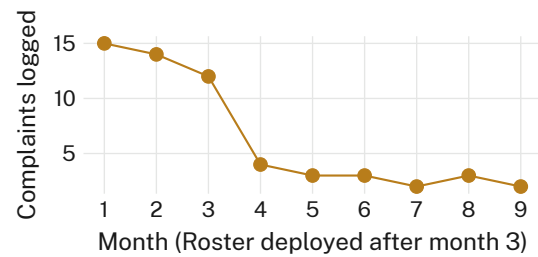


Figure 1: Kitchen-amenity fairness complaints logged per month fall from an average of 13.7 in the three months before the Reheat Slot Roster's deployment to 2.8 in the six months after — the number Facilities' continuation report cites as evidence the scheme worked.

Complaint volume (Figure 1) falls sharply and holds at the lower level: three pre-deployment months averaging 13.7 tickets give way to six post-deployment months averaging 2.8, a fall of just over four-fifths that shows no sign of reverting across either post-deployment quarter.

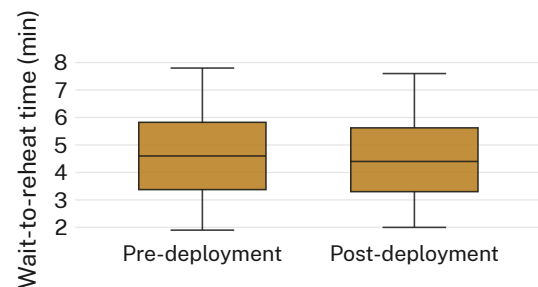


Figure 2: Wait-to-reheat time, stopwatched from queue arrival to reheat start, shows no detectable shift between matched pre-deployment ($n = 64$) and post-deployment ($n = 61$) lunch-hour samples — medians of 4.6 and 4.4 minutes, respectively.

The direct wait-time measurement (Figure 2) does not track the complaint count. Median wait-to-reheat time was 4.6 minutes before deployment and 4.4 minutes after, with heavily overlapping interquartile ranges and no significant difference by a two-sample rank test (n.s.). Staff are, on this evidence, waiting about as long to reheat their lunch as they did before the Roster existed; what has changed is how often they say so.

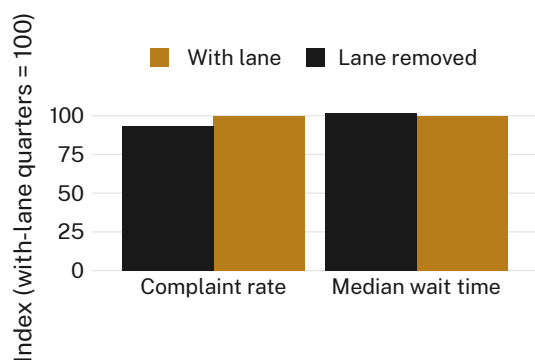


Figure 3: Withdrawing the Quick-Reheat Lane for a further quarter leaves both outcomes within the range already observed — indexed to the two prior quarters, complaint rate moves from 100 to 93 and median wait time from 100 to 102.

The ablation (Figure 3) changes almost nothing. Removing the Quick-Reheat Lane and returning to a uniform slot grid shifts the complaint rate by seven index points and median wait time by two, in opposite directions and both comfortably inside the quarter-to-quarter variation already present in the two prior quarters. We retain the finding rather than treat it as a design failure: a differentiated fast lane that neither measure can detect the absence of was, on this evidence, not doing detectable work in the first place.

Read together, the two outcomes move independently of one another across all three phases of the trial: complaint volume drops an order of magnitude at deployment and stays low through the ablation quarter, while wait-to-reheat time holds within half a minute of its pre-deployment level throughout, lane included or not.

Discussion

Two measures of the same intervention, recorded across the same population and calendar window, tell different stories. The complaint count says the Reheat Slot Roster worked; the wait-time measurement says the thing the complaint was nominally about — how long someone waits to reheat their lunch — has not moved. Both can be true at once. A published, visible mechanism gives staff a channel other than a complaint to route their frustration through, and a channel is enough to retire most of the complaints without retiring the underlying wait a fairness complaint is usually understood to index.

This is not evidence the Roster is worthless. A scheme that converts an informal, escalation-

prone queue into a legible, published roster may be doing real governance work — reducing interpersonal friction, giving line managers something to point to, converting a diffuse irritation into a bounded, visible artefact — without doing any of the queueing work its name implies. The gap sits comfortably alongside a broader pattern in which an institution's tracked measure and its substantive goal diverge under exactly the conditions this scheme meets: low cost, high visibility, and a service log built to count the measure rather than the goal.

Limitations

The trial covers one break room at one institution across three consecutive quarters; a busier appliance or a shorter lunch window might surface a Quick-Reheat effect this sample's ceiling cannot rule out, though the consistency of the null result across two independent measurement protocols and three trial phases suggests it is not an artefact of either method alone. Observers were not blind to the trial phase, which could bias a subjective assessment but not a stopwatched interval. The sample sizes ($n = 64, 61, 58$) are modest for a rank test, though the effect the test fails to detect would, at this sample size, need to be large enough that the continuation report's own complaint-count evidence should already have implied it.

Conclusion

A paper-slip reservation scheme can retire four-fifths of the complaints filed against a shared appliance without measurably changing how long anyone waits to use it. This paper finds exactly that pattern in Slop University's Reheat Slot Roster: a sharp, sustained fall in kitchen-amenity fairness complaints against a wait-to-reheat time that holds steady before, during, and after the scheme's one ablation. Facilities' continuation report is not wrong that the complaints stopped; it has simply not yet had reason to ask whether the wait did. Future work might instrument the appliance directly rather than relying on stopwatched sampling, a step this trial's budget did not extend to, to settle whether a longer window would find the effect this one could not.

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